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ABSTRACT

A recent publication presents a detailed prescription for constructing global eigenfunctions and complex growth rates for resistive instabilities in axisymmetric toroidal systems, using the method of matched asymptotic expansions. At each singular surface in the plasma, the small- x asymptotic solutions of the ideal MHD outer-region equations are matched to the large- x asymptotic solutions of the resistive inner-region equations, with x being the distance from the singular surface. The success of this method depends upon accurate asymptotic solutions of the resistive inner region equations. Extensive studies have shown that there are not only regimes of excellent agreement with straight-through methods, without asymptotic matching, but also regimes of strong disagreement. In an effort to find the cause of this discrepancy, we present here a different method of constructing asymptotic solutions of the inner region equations and we study the convergence properties of the new and old solutions.

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I. INTRODUCTION

A recent publication¹ presents a detailed prescription for constructing global eigenvalues and complex growth rates for resistive instabilities in axisymmetric toroidal systems, using the method of matched asymptotic expansions. At each singular surface in the plasma, the small- x asymptotic solutions of the ideal MHD outer-region equations are matched to the large- x asymptotic solutions of the resistive inner-region equations,² with x being the distance from the singular surface. The success of this method depends upon accurate asymptotic solutions of the resistive inner region equations.

Extensive studies have shown that the results of the method of matched asymptotic expansions have not only regimes of excellent agreement but also regimes of a serious discrepancy with those of a straight-through method without asymptotic matching.³ In an effort to find the cause of this discrepancy, we reconsider here the method used previously to construct asymptotic solutions to the inner region equations. We use a different method to construct solutions and examine the convergence properties of the old and new solutions.

The resistive inner region equations² constitute a 6th-order coupled system of ordinary differential equations (ODEs) with an irregular singular point as $x \rightarrow \infty$. There are 6 independent solutions: two exponentially large, two exponentially small, and two powerlike. The exponentially large solutions are excluded by boundary conditions; the

exponentially small solutions become negligible for large x ; and the powerlike solutions are matched to the corresponding terms in the outer region to construct the global solutions. So only the powerlike solutions are needed. The key issue is how to compute these solutions without interference from the exponentially large and small solutions.

The method used in Ref. 1, Sec. III, is a more compact but equivalent form of the method originally presented in Ref. 4, Appendix A. The equations are expressed as a system of three coupled “second”-order equations for the normal perturbed magnetic field Ψ , the normal perturbed plasma displacement Ξ , and the perturbed plasma pressure Υ . An ansatz is introduced into the equations consisting of a descending power series in $1/x^2$. This converts them to a hierarchy of algebraic equations to be solved order by order in $1/x^2$.

As a check on the validity of these older solutions, we consider a new treatment based on the method of Wasow, Chap. V.⁵ The equations are expressed in matrix form as a 6th-order system of coupled “first”-order equations. The leading-order coefficient matrix is transformed to Jordan canonical form, with the help of the Mathematica program.⁶ The diagonal elements include two positive eigenvalues for the large exponential solutions, two negative eigenvalues for the small exponential solutions, and two zero eigenvalues for the powerlike solutions.

A key result is that the two zero elements are coupled by an off diagonal 1, making the matrix nondiagonalizable. This makes all the

difference in the world. Many textbooks on ODEs treat solutions in the neighborhood of a singular point, but all except Wasow quit before reaching this most difficult case.

The remainder of this paper is organized as follows. Section II follows Wasow's procedure for constructing asymptotic solutions. Section III presents convergence studies of the asymptotic solutions and compares the old and new solutions. Section IV shows comparisons between the calculation of growth rates between the straight-through MARS-F code³ and resistive DCON using the ratio between the inner region $x_{\max}$ and the outer region $x_{\min}$ to verify overlap between the outer and inner regions. Section V presents discussion and conclusions.

II. ASYMPTOTIC SOLUTIONS

In this section, we solve the resistive inner region equations by the method of Wasow.⁵ In Subsection II A, we formulate the equations as a 6th-order coupled system of first-order ODEs in matrix form and transform the leading order coefficient matrix to Jordan canonical form. In Subsection II B, we present the form of an arbitrary linear change of dependent variables. In Subsection II C, we apply this to split the equations for the powerlike solutions from those for the large and small exponential solutions, which can be regarded as determining a Schur complement order by order. In Subsection II D, we further transform the reduced equation for the powerlike solutions. In Subsection II E, we introduce a shearing transformation to eliminate vanishing of the determinant of the zeroth-order equation. In Subsection II F, we solve the equations for the powerlike solutions. In Subsection II G, we assemble the full solution by back substitution with all of the transforming matrices used previously.

A. Jordan canonical form

The resistive inner region equations are²

$$\begin{aligned}\Psi_{xx} - H\Upsilon_x - Q(\Psi - x\Xi) &= 0, \\ Q^2\Xi_{xx} - Qx^2\Xi + Qx\Psi + (E+F)\Upsilon + H\Psi_x &= 0, \\ Q\Upsilon_{xx} - x^2\Upsilon + x\Psi + Q^2[G(\Xi - \Upsilon) - K(E\Xi + F\Upsilon + H\Psi_x)] &= 0,\end{aligned}\quad (1)$$

where the scaled dependent variables are Ψ , proportional to the perturbed normal magnetic field or the parallel vector potential; Ξ , proportional to the perturbed normal plasma displacement or the perturbed electrostatic potential; and Υ , proportional to the perturbed plasma pressure. The independent variable x is proportional to the distance from the singular surface. The parameters E , F , H , G , and K are dimensionless equilibrium quantities evaluated at the singular surface, defined in Ref. 2. Q is a scaled complex growth rate, proportional to the unscaled growth rate s , which is treated as a global eigenvalue. The first equation is derived from the parallel Ohm's law, the second from the quasineutrality equation, and the third from the adiabatic pressure law.

Equation (1) can be expressed in matrix form as a coupled system of first-order equations,

$$\Psi \equiv \begin{pmatrix} x\Psi \\ \Xi \\ \Upsilon \end{pmatrix}, \quad \mathbf{u} \equiv \begin{pmatrix} \Psi \\ \Psi'/x \end{pmatrix}, \quad \mathbf{u}' = x\mathbf{A}\mathbf{u}, \quad (2)$$

with the coefficient matrix

$$\mathbf{A} = \mathbf{A}_0 + x^{-2}\mathbf{A}_1 + x^{-4}\mathbf{A}_2, \quad (3)$$

$$\mathbf{A}_0 = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -Q & 0 & 0 & 0 & H \\ 0 & 1/Q & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/Q & 0 & 0 & 0 \end{pmatrix}, \quad (4)$$

$$\mathbf{A}_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 & 0 \\ H/Q^2 & 0 & 0 & 0 & 0 & 0 \\ -HKQ & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{A}_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ Q & 0 & 0 & 1 & 0 & 0 \\ -1/Q & 0 & -(E+F)/Q^2 & -H/Q^2 & -1 & 0 \\ -1/Q & -(G-KE)Q & (G+KF)Q & HKQ & 0 & -1 \end{pmatrix}. \quad (5)$$

Following Wasow, with help from Mathematica function Jordan decomposition,⁶ we transform $\mathbf{A}_0$ to Jordan canonical form,

$$\mathbf{u} = \mathbf{T}\mathbf{v}, \quad \mathbf{v}' = x\mathbf{J}\mathbf{v}, \quad \mathbf{J} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}, \quad \lambda \equiv Q^{-1/2}, \quad (6)$$

with

$$\mathbf{T} = \begin{pmatrix} 1 & 0 & HQ & Q^{5/2} & HQ & -Q^{5/2} \\ 0 & 0 & 0 & -Q^{1/2} & 0 & Q^{1/2} \\ 0 & 0 & -Q^{1/2} & 0 & Q^{1/2} & 0 \\ 0 & 1 & -HQ^{1/2} & -Q^2 & HQ^{1/2} & -Q^2 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad (7)$$

$$\mathbf{T}^{-1} = \begin{pmatrix} 1 & Q^2 & 0 & 0 & 0 & -HQ \\ 0 & 0 & -H & 1 & Q^2 & 0 \\ 0 & 0 & -1/(2Q^{1/2}) & 0 & 0 & 1/2 \\ 0 & -1/(2Q^{1/2}) & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 1/(2Q^{1/2}) & 0 & 0 & 1/2 \\ 0 & 1/(2Q^{1/2}) & 0 & 0 & 1/2 & 0 \end{pmatrix}, \quad (8)$$

$$\mathbf{J}_0 = \mathbf{T}^{-1}\mathbf{A}_0\mathbf{T} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\lambda & 0 & 0 & 0 \\ 0 & 0 & 0 & -\lambda & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda \end{pmatrix}. \quad (9)$$

We then transform the higher-order matrices,

$$\mathbf{J}_1 \equiv \mathbf{T}^{-1} \mathbf{A}_1 \mathbf{T}, \quad \mathbf{J}_2 \equiv \mathbf{T}^{-1} \mathbf{A}_2 \mathbf{T}. \quad (10)$$

The Jordan matrix $\mathbf{J}_0$ in Eq. (9) is nearly diagonal. There are two positive eigenvalues λ representing the large exponential solutions, two negative eigenvalues $-\lambda$ representing the small exponential solutions, and two zero eigenvalues representing the powerlike solutions. But there is a 1 above the diagonal in the second column, coupling the two powerlike solutions, which complicates the situation, making $\mathbf{J}_0$ nondiagonalizable.

B. Change of dependent variables

We rewrite the differential equation for a single column vector $\mathbf{v}$ to a fundamental matrix of solutions $\mathbf{V}$, with each column of $\mathbf{V}$ an independent solution of the differential equation, $\det \mathbf{V} \neq 0$, $\mathbf{V}' = x \mathbf{J} \mathbf{V}$. We introduce a change of dependent variables, which decouples the powerlike solutions from the large and small exponential solutions. The general form of a linear change of dependent variables is

$$\mathbf{V} = \mathbf{P} \mathbf{W}, \quad \det \mathbf{P} \neq 0, \quad \mathbf{W}' = x \mathbf{B} \mathbf{W}, \quad (11)$$

$$\mathbf{B} = \mathbf{P}^{-1} (\mathbf{J} \mathbf{P} - x^{-1} \mathbf{P}'). \quad (12)$$

The usual way such a transformation is used is to choose some $\mathbf{P}$ and explore the consequences. Instead, Wasow rewrites Eq. (12) as

$$\mathbf{P}' = x(\mathbf{J} \mathbf{P} - \mathbf{P} \mathbf{B}) \quad (13)$$

and treats it as an equation for determining a $\mathbf{P}$ with specified properties. We seek solutions in the form of descending power series

$$\begin{aligned} \mathbf{J} &= \sum_{k=0}^{\infty} \mathbf{J}_k x^{-2k}, \\ \mathbf{P} &= \sum_{k=0}^{\infty} \mathbf{P}_k x^{-2k}, \\ \mathbf{B} &= \sum_{k=0}^{\infty} \mathbf{B}_k x^{-2k}, \end{aligned} \quad (14)$$

with the understanding that $\mathbf{J}_k = 0$ for $k > 2$. Then, Eq. (13) becomes

$$\mathbf{J}_0 \mathbf{P}_0 - \mathbf{P}_0 \mathbf{B}_0 = 0, \quad (15)$$

$$\mathbf{J}_0 \mathbf{P}_k - \mathbf{P}_k \mathbf{B}_0 = \sum_{l=0}^{k-1} (\mathbf{P}_l \mathbf{B}_{k-l} - \mathbf{J}_{k-l} \mathbf{P}_l) - 2(k-1) \mathbf{P}_{k-1}, \quad k > 0. \quad (16)$$

C. Splitting transformation

We partition the space of solutions into two subspaces, the first 2-dimensional for the powerlike solutions and the second 4-dimensional for the large and small exponential solutions. We seek $\mathbf{P}$ such that $\mathbf{B}$ is block diagonal, with a 2×2 block $\mathbf{B}^{11}$ representing the powerlike solutions and a 4×4 block $\mathbf{B}^{22}$ representing the large and small exponential solutions. The vanishing of the off diagonal blocks $\mathbf{B}^{12}$ and $\mathbf{B}^{21}$ represents the decoupling we seek between the powerlike solutions and the exponential solutions. This can be regarded as determining the Schur complement order by order. While the original matrix $\mathbf{A}$ and the Jordan matrix $\mathbf{J}$ terminate after 3 terms, the transformed matrix $\mathbf{B}$ does not terminate.

Since the Jordan matrix $\mathbf{J}_0$ has the desired block form, we can write

$$\mathbf{J}_0 = \begin{pmatrix} \mathbf{J}_0^{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_0^{22} \end{pmatrix}, \quad \mathbf{B}_0 = \mathbf{J}_0, \quad \mathbf{P}_0 = \mathbf{I}. \quad (17)$$

Then, Eq. (16) becomes

$$\mathbf{J}_0 \mathbf{P}_k - \mathbf{P}_k \mathbf{J}_0 = \mathbf{B}_k - \mathbf{K}_k, \quad k > 0, \quad (18)$$

with

$$\mathbf{K}_k \equiv \mathbf{J}_k + 2(k-1) \mathbf{P}_{k-1} + \sum_{l=1}^{k-1} (\mathbf{J}_{k-l} \mathbf{P}_l - \mathbf{P}_l \mathbf{B}_{k-l}). \quad (19)$$

Using the block decompositions

$$\begin{aligned} \mathbf{B}_k &= \begin{pmatrix} \mathbf{B}_k^{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_k^{22} \end{pmatrix}, \\ \mathbf{P}_k &= \begin{pmatrix} \mathbf{0} & \mathbf{P}_k^{12} \\ \mathbf{P}_k^{21} & \mathbf{0} \end{pmatrix}, \\ \mathbf{K}_k &= \begin{pmatrix} \mathbf{K}_k^{11} & \mathbf{K}_k^{12} \\ \mathbf{K}_k^{21} & \mathbf{K}_k^{22} \end{pmatrix}, \end{aligned} \quad (20)$$

we can solve Eq. (18),

$$\mathbf{B}_k^{11} = \mathbf{K}_k^{11}, \quad \mathbf{B}_k^{22} = \mathbf{K}_k^{22}, \quad (21)$$

$$\begin{aligned} \mathbf{J}_0^{22} \mathbf{P}_k^{21} - \mathbf{P}_k^{21} \mathbf{J}_0^{11} &= -\mathbf{K}_k^{21}, \\ \mathbf{J}_0^{11} \mathbf{P}_k^{12} - \mathbf{P}_k^{12} \mathbf{J}_0^{22} &= -\mathbf{K}_k^{12}. \end{aligned} \quad (22)$$

Equation (22) is a Lyapunov equation, which can be solved algebraically by Mathematica function `LyapunovSolve` or numerically for $\mathbf{P}_k$ in terms of known quantities determined in the previous iterations. The first few terms of $\mathbf{B}^{11}$ are

$$\begin{aligned} \mathbf{B}_0^{11} &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \\ \mathbf{B}_1^{11} &= \begin{pmatrix} H & -KH^2Q^2 \\ 0 & 1-H \end{pmatrix}, \\ \mathbf{B}_2^{11} &= \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, \end{aligned} \quad (23)$$

with

$$\begin{aligned} b_{11} &= -HQ^2[1 + K(E + F - H + H^2)], \\ b_{12} &= HQ\{1 - HK[E - Q^3(1 + FK + H^2K)] \\ &\quad + GH(1 + KQ^3)\}, \\ b_{21} &= 2(H - 1) - (E + F + H^2), \\ b_{22} &= Q^2\{H[1 + K(E + F - H + H^2)] - 2\}. \end{aligned} \quad (24)$$

D. Transformation of the reduced equation

In this section, we study the reduced equation for the powerlike solutions, having decoupled it from the large and small exponential

solutions. All the matrices involved are 2×2 , the upper-left blocks of the original 6×6 matrices. Hereafter, we suppress the superscripts.

The equation to be solved is

$$\mathbf{W}' = x\mathbf{B}\mathbf{W}, \quad \mathbf{B} = \sum_{k=0}^{\infty} \mathbf{B}_k x^{-2k}, \quad \mathbf{B}_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (25)$$

Following Wasow, we again change dependent variables, this time with a matrix $\mathbf{Q}$,

$$\mathbf{W} = \mathbf{Q}\mathbf{X}, \quad \det \mathbf{Q} \neq 0, \quad \mathbf{X}' = x\mathbf{C}\mathbf{X}, \quad (26)$$

$$\mathbf{C} = \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q} - x^{-1}\mathbf{Q}^{-1}\mathbf{Q}', \quad \mathbf{Q}' = x(\mathbf{B}\mathbf{Q} - \mathbf{Q}\mathbf{C}). \quad (27)$$

As before, we seek descending power series solutions for $\mathbf{Q}$ and $\mathbf{C}$,

$$\mathbf{Q} = \sum_{k=0}^{\infty} \mathbf{Q}_k x^{-2k}, \quad \mathbf{C} = \sum_{k=0}^{\infty} \mathbf{C}_k x^{-2k}, \quad (28)$$

$$\mathbf{Q}_0 = \mathbf{I}, \quad \mathbf{C}_0 = \mathbf{B}_0. \quad (29)$$

This transforms Eq. (27) into

$$\mathbf{B}_0\mathbf{Q}_0 - \mathbf{Q}_0\mathbf{B}_0 = 0, \quad \mathbf{Q}_0 = \mathbf{I}, \quad (30)$$

$$\mathbf{B}_0\mathbf{Q}_k - \mathbf{Q}_k\mathbf{B}_0 = \mathbf{C}_k - \mathbf{K}_k, \quad k > 0, \quad (31)$$

with

$$\begin{aligned} \mathbf{K}_k &\equiv \mathbf{B}_k + 2(k-1)\mathbf{Q}_{k-1} + \sum_{l=1}^{k-1} (\mathbf{B}_{k-l}\mathbf{Q}_l - \mathbf{Q}_l\mathbf{C}_{k-l}) \\ &= \begin{pmatrix} k_k^{11} & k_k^{12} \\ k_k^{21} & k_k^{22} \end{pmatrix}. \end{aligned} \quad (32)$$

For arbitrary matrices

$$\mathbf{Q} = \begin{pmatrix} q^{11} & q^{12} \\ q^{21} & q^{22} \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} r^{11} & r^{12} \\ r^{21} & r^{22} \end{pmatrix}, \quad (33)$$

using Eq. (25) for $\mathbf{B}_0$, an inhomogeneous equation of the form Eq. (31),

$$\mathbf{B}_0\mathbf{Q} - \mathbf{Q}\mathbf{B}_0 = \mathbf{R}, \quad (34)$$

can be written in terms of components as

$$\begin{aligned} q^{21} &= r^{11}, \quad q^{22} - q^{11} = r^{12}, \\ r^{21} &= 0, \quad r^{11} + r^{22} = 0. \end{aligned} \quad (35)$$

Thus, there are two solutions for the components of $\mathbf{Q}$ and two constraints on the components of $\mathbf{R}$. Since there are 4 conditions on 8 variables, four of the quantities are undetermined and arbitrary. Following Wasow, we choose the upper two components of $\mathbf{Q}$ and $\mathbf{R}$ to vanish and use the lower two components to satisfy Eq. (31),

$$\mathbf{Q}_k = \begin{pmatrix} 0 & 0 \\ -k_k^{11} & -k_k^{12} \end{pmatrix}, \quad \mathbf{C}_k = \begin{pmatrix} 0 & 0 \\ -k_k^{21} & -k_k^{22} \end{pmatrix}. \quad (36)$$

E. Shearing transformation

The reduced equation now has the form

$$\mathbf{X}' = x\mathbf{C}\mathbf{X}, \quad \mathbf{C} = \sum_{k=0}^{\infty} \mathbf{C}_k x^{-2k}, \quad (37)$$

with

$$\begin{aligned} \mathbf{C}_0 &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad \mathbf{C}_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}; \\ \mathbf{C}_k &= \begin{pmatrix} 0 & 0 \\ c_k^{21} & c_k^{22} \end{pmatrix}, \quad k > 1, \end{aligned} \quad (38)$$

and, in particular,

$$c_2^{21} = -(E + F + H + 2). \quad (39)$$

The leading-order matrix $\mathbf{C}_0$ is singular. To eliminate this problem, we introduce a shearing transformation,

$$\mathbf{X} = \mathbf{S}\mathbf{Y}, \quad x\mathbf{Y}' = \mathbf{D}\mathbf{Y}, \quad (40)$$

with

$$\mathbf{S} = \begin{pmatrix} 1 & 0 \\ 0 & 1/x^2 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & x^2 \end{pmatrix}, \quad \mathbf{S}' = \begin{pmatrix} 0 & 0 \\ 0 & -2/x^3 \end{pmatrix}, \quad (41)$$

and

$$\mathbf{D} = x\mathbf{S}^{-1}(x\mathbf{C}\mathbf{S} - \mathbf{S}') = \sum_{k=0}^{\infty} \mathbf{D}_k x^{-2k}, \quad (42)$$

$$\mathbf{D}_0 = \begin{pmatrix} 0 & 1 \\ c_2^{21} & 3 \end{pmatrix}; \quad \mathbf{D}_k = \begin{pmatrix} 0 & 0 \\ c_{k+2}^{21} & c_{k+1}^{22} \end{pmatrix}, \quad k > 0. \quad (43)$$

F. Solution of the reduced equation

We can now solve the reduced equation for $\mathbf{Y}$, Eq. (40). We seek a descending power series solution of the form

$$\begin{aligned} \mathbf{Y} &= \sum_{k=0}^{\infty} \mathbf{Y}_k x^{\mathbf{R}-2kl}, \\ x\mathbf{Y}' &= \sum_{k=0}^{\infty} \mathbf{Y}_k (\mathbf{R} - 2kl) x^{\mathbf{R}-2kl}, \\ \mathbf{D}\mathbf{Y} &= \sum_{l=0}^{\infty} \mathbf{D}_l \sum_{k=0}^{\infty} \mathbf{Y}_k x^{\mathbf{R}-2(k+l)l}, \\ &= \sum_{k=0}^{\infty} \sum_{l=0}^k \mathbf{D}_l \mathbf{Y}_{k-l} x^{\mathbf{R}-2kl}, \end{aligned} \quad (44)$$

with

$$\mathbf{R} = \begin{pmatrix} r_+ & 0 \\ 0 & r_- \end{pmatrix}, \quad (45)$$

a diagonal matrix of nonintegral powers, to be determined by an eigenvalue equation. This converts Eq. (40) to

$$\sum_{k=0}^{\infty} \left[\mathbf{Y}_k (\mathbf{R} - 2kl) - \sum_{l=0}^k \mathbf{D}_l \mathbf{Y}_{k-l} \right] x^{\mathbf{R}-2kl} = 0. \quad (46)$$

Solving this order by order, we obtain

$$\mathbf{Y}_k(\mathbf{R} - 2k\mathbf{I}) - \mathbf{D}_0\mathbf{Y}_k = \sum_{l=1}^k \mathbf{D}_l\mathbf{Y}_{k-l}. \quad (47)$$

To the lowest order,

$$\mathbf{Y}_0\mathbf{R} - \mathbf{D}_0\mathbf{Y}_0 = 0, \quad \mathbf{Y}_0 = \begin{pmatrix} 1 & 1 \\ r_+ & r_- \end{pmatrix}, \quad (48)$$

with

$$r_{\pm} = \frac{3}{2} \pm \sqrt{-D_I}, \quad D_I = E + F + H - \frac{1}{4}. \quad (49)$$

These are the well-known Mercier powers, required to match to the ideal outer region, except for the term $3/2$ instead of the usual $-1/2$. This discrepancy is resolved by back substitution, in Sec. II G, where it is found that the zeroth-order term in the normal displacement Ξ vanishes and the next term reduces the power by 2.

To solve the higher-order equations for $\mathbf{Y}_k$, we make one more transformation,

$$\mathbf{Y} = \mathbf{Y}_0\mathbf{Z}, \quad \mathbf{Z} = \sum_{k=0}^{\infty} \mathbf{Z}_k x^{\mathbf{R}-2k\mathbf{I}}, \quad \mathbf{Z}_0 = \mathbf{I}, \quad (50)$$

$$\mathbf{Y}_0^{-1}\mathbf{D}_0\mathbf{Y}_0 = \mathbf{R}, \quad \mathbf{E}_k \equiv \mathbf{Y}_0^{-1}\mathbf{D}_k\mathbf{Y}_0, \quad (51)$$

$$\mathbf{Z}_k(\mathbf{R} - 2k\mathbf{I}) - \mathbf{R}\mathbf{Z}_k = \sum_{l=1}^k \mathbf{E}_l\mathbf{Z}_{k-l}, \quad k > 0. \quad (52)$$

Equation (52) is easily solved numerically.

G. Back substitution

To recover the physical variables in Eq. (2), we perform back substitution with all the transformation matrices we have introduced. In order of appearance, they are the 6×6 Jordan transformation matrix $\mathbf{T}$, Eq. (7); the 6×6 splitting transformation $\mathbf{P}$, Eq. (13); the 2×2 reduced transformation $\mathbf{Q}$, Eq. (33); and the 2×2 shearing transformation matrix $\mathbf{S}$, Eq. (41). Since we only compute the powerlike solutions and not the large and small exponential solutions, we only need the 6×2 part of the $\mathbf{P}$ matrix. The result is

$$\mathbf{U} = \mathbf{TPQSY}, \quad (53)$$

where $\mathbf{U}$ is a 6×2 matrix of solutions, the first column corresponding to r_+ and the second to r_- . This completes the construction of the asymptotic solutions.

III. CONVERGENCE TESTS

We have implemented the asymptotic solutions in a Fortran 95 code INPS. It computes the matrices in Eq. (53) and then returns the solutions $\mathbf{U}$ and their derivatives at arbitrary input variable x .

To test the convergence of these solutions to solutions of the differential equations, we define and evaluate

$$\Delta_{\pm} \equiv \frac{\|\mathbf{u}'_{\pm} - x\mathbf{A}\mathbf{u}_{\pm}\|}{\max(\|\mathbf{u}'_{\pm}\|, \|x\mathbf{A}\mathbf{u}_{\pm}\|)}, \quad (54)$$

a measure of the error in solving the differential equation, with $\|\cdot\|$ denoting the maximum norm, for each of the two Mercier powers.

Figure 1 shows a graph of $\log_{10}\Delta_+$ vs $\log_{10}x$, using parameters

$$\begin{aligned} E &= -3.369 \times 10^{-2}, & F &= 2.409 \times 10^{-3}, \\ H &= 1.292 \times 10^{-2}, & G &= 8.950 \times 10^1, \\ K &= 2.332 \times 10^2, & Q &= 1.234 \times 10^{-1}, \end{aligned} \quad (55)$$

appropriate to a simulated D-shaped equilibrium with aspect ratio 2, evaluated at the $q=2$ surface, showing the effect of $k_{\max}$, the highest-order term in the power series, on the rate of convergence. This constitutes a verification of the accuracy of the asymptotic solutions. Graphs of Δ_- have similar behavior.

Figure 2 shows a graph of convergence with the same parameters except with Q , the scaled growth rate, increasing logarithmically from 10^{-2} to 10^{+2} and with $k_{\max} = 8$, showing the effect of increasing Q on the rate of convergence.

Figure 3 shows the dependence of the cutoff position $x_{\max}$ as a function of the scaled growth rate Q . This choice of $x_{\max}$ is crucial to the ability of the method of matched asymptotic expansions to compute global eigenfunctions and growth rates. If $x_{\max}$ is too large, it causes a failure of overlap of the inner and outer regions.

Figure 4 shows that the old and new solutions are asymptotically equivalent. This was somewhat unexpected, since the derivations and the forms of the solutions are so different. It implies that the same procedure for determining $x_{\max}$ applies to the old solution as well.

Thus, while the new solutions cannot be expected to solve the problem of discrepancy with the straight-through method any better than the old ones, improved understanding of the convergence properties of both solutions, as obtained here, and the new and more reliable method for choosing $x_{\max}$, should contribute to the quest for a resolution.

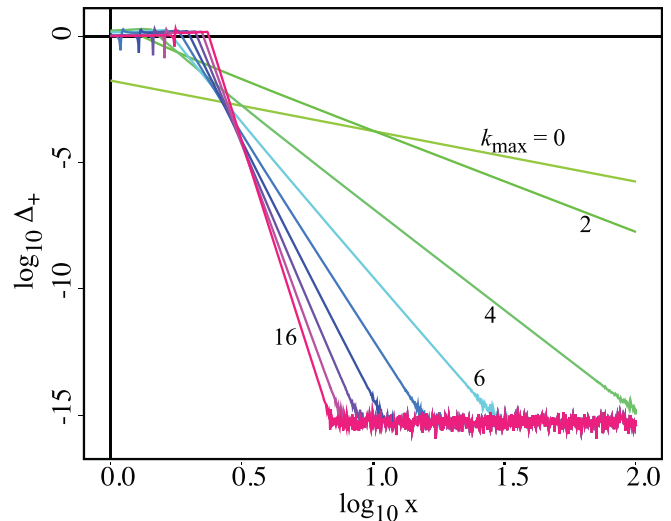


FIG. 1. The effect of $k_{\max}$ on convergence of the asymptotic solution to a solution of the differential equation, $\log_{10}\Delta_+$ vs $\log_{10}x$. Straight lines indicate powerlike convergence. Numbers denote $k_{\max}$, the highest term in the power series. Increasing negative slopes indicate convergence of $\Delta_{\pm} \sim x^{-2k_{\max}}$, $k_{\max} > 1$. Jagged lines at the bottom indicate roundoff.

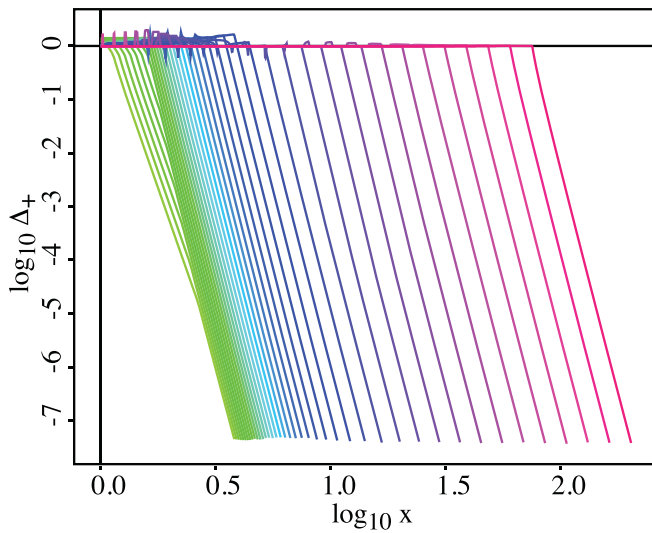


FIG. 2. The effect of scaled growth rate Q on convergence of the asymptotic solution to a solution of the differential equation, $\log_{10} \Delta_+$ vs $\log_{10} x$, with $k_{\max} = 8$. Different curves denote logarithmically increasing values of Q from 10^{-2} to 10^{+2} . The effect of increasing Q is to increase the value of $x = x_{\text{crit}}$ at which powerlike convergence starts. Each curve is terminated when $\Delta_{\pm} < \epsilon = 10^{-7}$, as a means of determining the optimal value of $x = x_{\max}$ for cutoff.

IV. COMPARISONS OF ASYMPTOTIC MATCHING AND STRAIGHT-THROUGH CALCULATIONS

A key motivation for this work was a failure of overlap between the inner and outer solution asymptotic regions, as discussed in Secs. III and IV of Ref. 1. With the improved estimates of $x_{\max}$ obtained in Sec. III, together with estimates of the outer region $x_{\min}$ based on similar methods applied to the asymptotic solutions of Sec. II of Ref. 1, we define $x_{\text{ratio}} \equiv x_{\min}/x_{\max}$. A good measure of overlap is then $x_{\text{ratio}} > 1$.

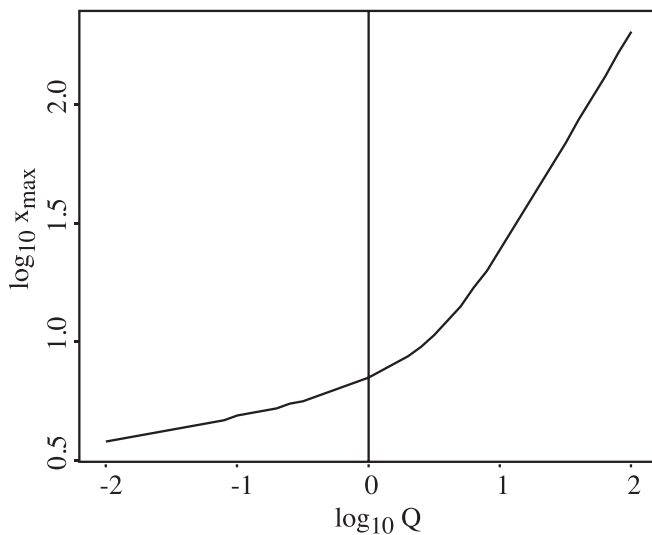


FIG. 3. $\log_{10} x_{\max}$ vs $\log_{10} Q$. The value of $x_{\max}$ is chosen as the position at which the error in Fig. 2 drops to 10^{-7} .

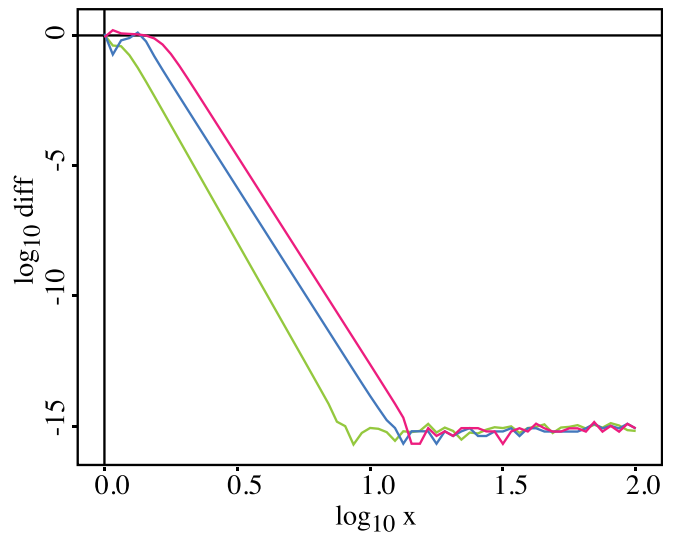


FIG. 4. Asymptotic equivalence, $\log_{10}$ of the relative difference between the old and new solutions for Ψ (green), Ξ (blue), and Υ (red) vs $\log_{10} x$, converging asymptotically to zero as $x^{-2k_{\max}}$, with $k_{\max} = 8$.

Figure 5 shows a comparison between growth rates computed by the MARS-F straight-through code³ and resistive DCON code using asymptotic matching, vs normalized β as it approaches the ideal MHD no-wall limit. This is for an equilibrium with circular cross section, aspect ratio 2, one singular surface at $q = 2$, and Lundquist number

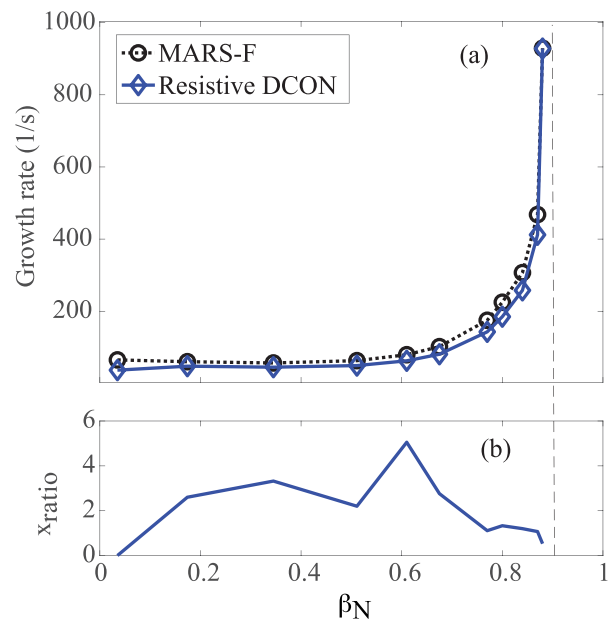


FIG. 5. (a) Comparison between growth rates computed by MARS-F and Resistive DCON vs normalized β_N , approaching ideal no-wall limit at $\beta_N = 1$. (b) x_{ratio} , the ratio of the outer region $x_{\min}$ to the inner region $x_{\max}$; $x_{\text{ratio}} > 1$ indicates good overlap between the outer and inner region solutions. Markers indicate the computed values, curves are straight lines connecting the markers.

$S = 10^8$. The bottom figure shows the overlap parameter x_{ratio} as a function of β_N . Since x_{ratio} scales as $S^{1/3}$, larger values of S give improved overlap.

The pressure scan indicates that the tearing mode can be driven while increasing the β value approaching the β no wall limit, though the Glasser effect in the inner layer provides the stabilizing effect. This result shows that it is essential to evaluate the tearing instability including both the inner and the outer regions.

Figure 6 shows a comparison between growth rates computed by MARS-F and resistive DCON code vs triangularity τ as it approaches the ideal MHD no-wall limit. This is for the same equilibrium as for Fig. 5 except that $\beta_N = 0.875$ and elongation $\kappa = 1.6$. The bottom figure shows the overlap parameter x_{ratio} as a function of τ .

Figure 7 shows a comparison between growth rates computed by MARS-F and resistive DCON code vs elongation κ . This is for the same equilibrium as for Fig. 6 except that $\tau = 0$. The bottom figure shows the overlap parameter x_{ratio} as a function of β_N .

While these figures indicate much improved agreement between DCON and MARS-F, the agreement is not perfect. We have not determined the cause of the remaining discrepancies or which code can be held responsible.

Elongation κ and triangularity τ have significant impacts on the tearing instability, which indicates the importance of solving the tearing instability in the full toroidal geometry to better understand the behavior the tearing instability. Here, increasing both elongation and triangularity can destabilize the tearing instability.

A more thorough comparison between resistive DCON and MARS-F, including equilibria with multiple singular surfaces, will be treated in a later publication.

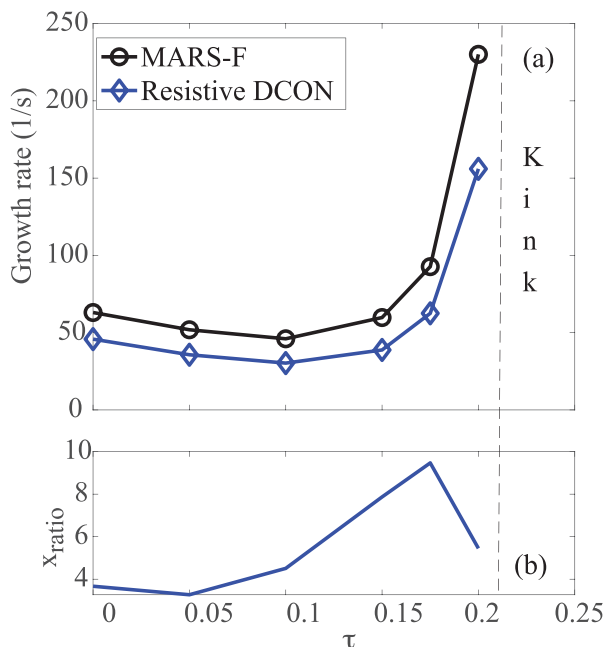


FIG. 6. (a) Comparison between growth rates computed by MARS-F and Resistive DCON vs triangularity τ , approaching ideal no-wall limit at $\tau = 0.21$. (b) x_{ratio} , the ratio of the outer region x_{min} to the inner region x_{max} .

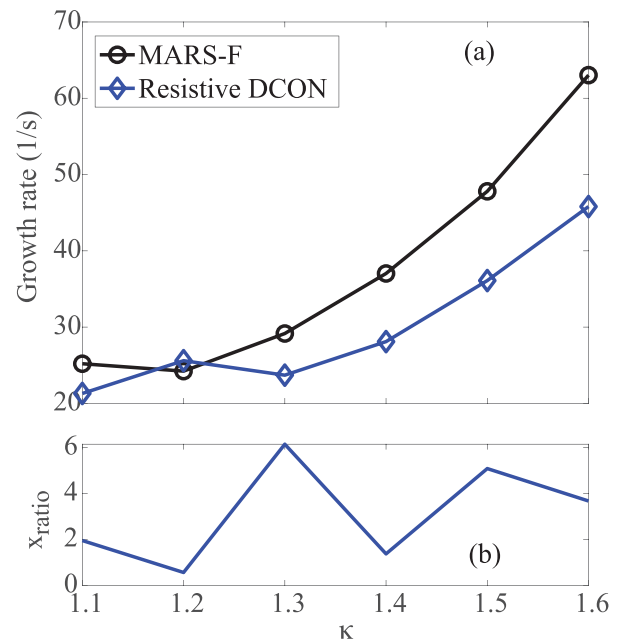


FIG. 7. (a) Comparison between growth rates computed by MARS-F and Resistive DCON vs elongation κ . (b) x_{ratio} , the ratio of the outer region x_{min} to the inner region x_{max} .

V. DISCUSSION AND CONCLUSIONS

We have derived an alternative asymptotic solution to the resistive tearing inner region equations. The new method, based on Wasow's treatment,⁵ identifies and corrects two apparent problems in the previous approach: determination of the correct Schur complement to decouple the powerlike solutions from the exponential solutions, using the splitting transformation; and the degeneracy of the Jordan canonical form, using the shearing transformation. We have demonstrated strong convergence to the solution of the differential equation. This provides a more reliable determination of the cutoff position x_{max} for the finite difference computation of the resistive region matching conditions and improved agreement of the full resistive DCON code with the straight-through MARS code. The new solution is unexpectedly asymptotically equivalent to the old solution.

Using the new cutoff position x_{max} and an improved Newton iteration, we have demonstrated improved agreement between resistive DCON and the MARS-F code, as discussed in Sec. IV.

The method developed here should also be applicable to more advanced inner region dynamics.^{7–9} This will be investigated in future studies.

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